

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

File extension for a file name is usually imposed by modern file systems (like NTFS and Ext4).

Select one:

☐ True

☒ False ✓

Question 2

Correct

Mark 1.00 out of 1.00

Flag question

A semaphore is a **synchronization** mechanisms that allows **more** threads be at one moment in the **critical region** protected by that semaphore. It provides the **P()** and **V()** primitives.

Your answer is correct.

Question 3

Correct

Mark 1.00 out of 1.00

Flag question

From the user perspective, a *text file* is a file in which every **byte** is a code of a printable **character** and also contain special codes for marking the **end** of a text **line**.

Question 4

Correct

Mark 1.00 out of 1.00

Flag question

The two main components of a file in any file systems are:

Select one or more:

☒ a. the file's metadata structure ✓

☐ b. file's owner

☒ c. the file's data (i.e. contents) ✓

☐ d. file's access permissions

☐ e. file's parent directory

Your answer is correct.

Question 5

23 ✓ 24 ✓

Show one page at a time

Finish review

Question 5

Correct

Mark 1.00 out of 1.00

Flag question

You are given the following code, which creates a file. Supposing the file is created successfully, which of the following actions can be performed successfully on the created file.

```
int fd = create("test", 0651);
```

Select one or more:

- ☒ a. owner can read the file ✓
- ☒ b. owner can write the file ✓
- ☐ c. owner can execute the file
- ☒ d. users in group with owner can read the file ✓
- ☐ e. users in group with owner can write the file
- ☒ f. users in group with owner can execute the file ✓
- ☐ g. other users can read the file
- ☐ h. other users can write the file
- ☒ i. other users can execute the file ✓

Your answer is correct.

Question 6

Correct

Mark 1.00 out of 1.00

Flag question

Supposing that a computer has 57 (logical) processors, how many threads could be simultaneously in the *running* state?

Answer: 57 ✓

Question 7

Correct

Mark 1.00 out of 1.00

Flag question

One reason an operating system could easily be ported on another hardware architecture is that each processor supports each own set of instructions.

Select one:

- ☐ True
- ☒ False ✓

Question 8

How many threads could run in the same time in the critical region of the function `aeronlane()` in the code below?



question

☐ False ✓

Question 8

Correct

Mark 1.00 out of 1.00

Flag question

How many threads could run in the same time in the critical region of the function `aeroplane()` in the code below?

```
pthread_mutex_t airport = PTHREAD_MUTEX_INITIALIZER;

void* aeroplane(void* arg)
{
    pthread_mutex_lock(&airport);

    // use a runway to take off or land

    pthread_mutex_unlock(&airport);
}

int main(int argc, char **argv)
{
    pthread_t th[23];

    for (i = 0; i < 23; i++) {
        pthread_create(&th[i], NULL, aeroplane, NULL);
    }

    for (i = 0; i < 23; i++) {
        pthread_join(th[i], NULL);
    }
}
```

Answer: 1 ✓

Question 9

Correct

Mark 1.00 out of 1.00

Flag question

Establish for the following file paths their correct type.

`"/home/os/file"` is a(n) absolute path

`"../os/file"` is a(n) relative path

`"file"` is a(n) relative path

`"./home/os/file"` is a(n) relative path

`"/"` is a(n) absolute path

Your answer is correct.

Question **10**

Correct

Mark 1.00 out of 1.00

Flag question

From a *logical perspective*, the OS position in a computing system's configuration is **above** the hardware and **below** the user applications.

Your answer is correct.

Question **11**

Correct

Mark 1.00 out of 1.00

Flag question

How many threads could run in the same time in the critical region of the function `aeroplane()` in the code below?

```
sem_t airport;

void* aeroplane(void* arg)
{
    sem_wait(&airport);

    // use a runway to take off or land

    sem_post(&airport);
}

int main(int argc, char **argv)
{
    pthread_t th[43];

    sem_init(&airport, 0, 8);

    for (i = 0; i < 43; i++) {
        pthread_create(&th[i], NULL, aeroplane, NULL);
    }

    for (i = 0; i < 43; i++) {
        pthread_join(th[i], NULL);
    }
}
```

Answer: ✓

Question **12**

Partially correct

Mark 0.30 out of 1.00

Flag question

Which memory management strategies use base and limit registers for translating virtual addresses to physical addresses?

Select one or more:

- ☒ a. segmentation ✓
- ☐ b. paging
- ☐ c. single contiguous area for a process
- ☐ d. best fit

Question **12**

Partially correct

Mark 0.30 out of 1.00

Flag question

Which memory management strategies use base and limit registers for translating virtual addresses to physical addresses?

Select one or more:

- ☒ a. segmentation ✓
- ☐ b. paging
- ☐ c. single contiguous area for a process
- ☐ d. best fit
- ☒ e. swapping ✗

Your answer is partially correct.

You have correctly selected 1.

Question **13**

Partially correct

Mark 0.83 out of 1.00

Flag question

You are given the following code of a program creating multiple threads. Identify which threads are CPU-bound and which are I/O bound.

```
#define N 6

void* fc_1(void* arg)
{
    long int steps = (long int) arg;
    int res = 0;

    for (int i = 0; i < 100 * steps; i++) {
        res += pow(res, i) + sqrt(i) * sqrt(i);
    }

    int *val = (int*) malloc(sizeof(int));
    *val = res;
    return val;
}

void* fc_2(void* arg)
{
    long int steps = (long int) arg;
    int res = 0;

    for (int i = 0; i < 1000 * steps; i++) {
        printf("i = %d\n", i);
        res += res + i;
        printf(res = %d\n", res);
    }

    return NULL;
}

int main(void)
{
    pthread th[N];
```

```
return NULL,
}

int main(void)
{
    pthread th[N];

    for(long int i = 0; i < N; i++) {
        if (i % 2 == 0) {
            pthread_create(&th[i], NULL, fc_2, (void*) i);
        } else {
            pthread_create(&th[i], NULL, fc_1, (void*) i);
        }
    }

    for (long int i = 0; i < N; i++) {
        pthread_join(th[i], NULL);
    }

    return 0;
}
```

th[5] CPU-bound ✓

th[4] CPU-bound ✗

th[3] CPU-bound ✓

th[0] I/O-bound ✓

th[1] CPU-bound ✓

th[2] I/O-bound ✓

Your answer is partially correct.
You have correctly selected 5.

Question 14

Correct

Mark 1.00 out of 1.00

Flag question

Match the number of processes and anonymous pipes created by the shell in order to handle the following command line.

```
ls -l /etc | cat | grep -i "os-exam" | sort
```

no of processes 4 ✓

no of pipes 3 ✓

Your answer is correct.

Question **15**

Correct

Mark 1.00 out of 1.00

Flag question

The two main components of a file in any file systems are:

Select one or more:

- ☐ a. file's parent directory
- ☐ b. file's access permissions
- ☐ c. file's owner
- ☒ d. the file's metadata structure ✓
- ☒ e. the file's data (i.e. contents) ✓

Your answer is correct.

Question **16**

Correct

Mark 1.00 out of 1.00

Flag question

A fundamental hardware support an OS uses to protect itself and the hardware from user processes is:

Select one:

- ☐ a. different disks used to store the OS and the user applications, respectively, the OS on the encrypted disk, while the user applications on the unencrypted ones
- ☒ b. a hierarchy of CPU execution modes with different privileges, the OS running in the most privileged one, while the user applications in the less privileged ones ✓
- ☐ c. different processors used to run the OS and user applications, respectively, the OS running on the boot-strap processor, while the user applications on the application processors
- ☐ d. different memory chips used to store the OS and the user applications, respectively, the OS being loaded on the fastest one, while the user applications on the slower ones

Your answer is correct.

Question **17**

Correct

Mark 1.00 out of 1.00

Flag question

Which types of pipes do the Linux and Windows operating systems provide?

Select one or more:

- ☐ a. unidirectional
- ☒ b. anonymous ✓
- ☒ c. named ✓
- ☐ d. reader-writer
- ☐ e. triangular

Your answer is correct.

Question **18**

Correct

Mark 1.00 out of 1.00

Flag question

An operating system is closely tied to (i.e. dependent by) the hardware it runs on.

Select one:

- ☒ True ✓
- ☐ False

Question **19**

Correct

Mark 1.00 out of 1.00

Flag question

Which of the following could be valid process states during a process lifetime?

Select one or more:

- ☐ a. jaberring
- ☒ b. blocked ✓
- ☐ c. jumping
- ☐ d. flattering
- ☒ e. terminated (like the Linux zombie state) ✓
- ☒ f. running ✓
- ☒ g. ready ✓

Your answer is correct.

Question **20**

Correct

Mark 1.00 out of 1.00

Flag question

An absolute path in Linux always starts with a .

Your answer is correct.

Question **21**

Correct

Mark 1.00 out of 1.00

Flag question

Which is the size in bytes of a process virtual address space in a system using 4 bits for representing a virtual memory address? You must give a number, not a mathematical formula.

Answer: ✓